

Arsenic Trioxide Cardiotoxicity Prediction Report

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I. Background

Arsenic Trioxide (ATO) is a first-line drug for treating Acute Promyelocytic Leukemia (APL). However, ATO may cause cardiotoxicity during treatment, including QT interval prolongation and arrhythmias, which can lead to fatal cardiac events in severe cases. Arsenic undergoes a series of methylation reactions in the body, and the distribution of its metabolites is closely related to the occurrence of cardiotoxicity.

This report uses a deep learning model to comprehensively analyze the patient's arsenic metabolism indicators (iAs, MMA, DMA), metabolic enzyme activity indices (PMI, SMI), and concomitant medication status to predict the risk of ATO cardiotoxicity and provide scientific evidence for clinical decision-making.

II. Test Data

2.1 Input Parameters

Indicator	Value	Unit	Description
iAs	50.00	ng/mL	Inorganic arsenic concentration
MMA	100.00	ng/mL	Monomethylarsonic acid concentration
DMA	100.00	ng/mL	Dimethylarsinic acid concentration
Cardiotoxic Drug	Yes	-	Concurrent use of other cardiotoxic drugs

2.2 Calculated Parameters

Indicator	Value	Percentage	Description
tAs	250.00	100.0%	Total arsenic = iAs + MMA + DMA
iAs%	50.00	20.0%	Inorganic arsenic percentage
MMA%	100.00	40.0%	Monomethylarsonic acid percentage

Indicator	Value	Percentage	Description
DMA%	100.00	40.0%	Dimethylarsinic acid percentage
PMI	2.000	-	Primary methylation index = MMA / iAs
SMI	1.000	-	Secondary methylation index = DMA / MMA

III. Prediction Results

Classification:	Positive (Toxic)
Toxicity Probability:	90.8%
Risk Level:	High Risk

Interpretation: This patient has a high risk of ATO cardiotoxicity. Close ECG monitoring and treatment adjustment are recommended.

IV. Feature Importance Analysis (SHAP Values)

SHAP (SHapley Additive exPlanations) values explain machine learning model predictions. Positive values indicate the feature increases toxicity risk, while negative values indicate the feature decreases risk. The larger the absolute value, the more significant the feature's impact.

Feature	SHAP Value	Impact Direction	Rank
tAs (Total Arsenic)	0.4089	Increase Risk	1
SMI (Secondary Methylation Index)	0.0540	Increase Risk	2
MMA% (MMA Percentage)	0.0448	Increase Risk	3
DMA% (DMA Percentage)	0.0215	Increase Risk	4
CT_drug (Cardiotoxic Drug)	-0.0657	Decrease Risk	5

Major Risk Factor: tAs (Total Arsenic Concentration)

V. Clinical Recommendations

Based on the prediction results and feature analysis, the following clinical recommendations are provided:

- **High Total Arsenic Concentration:** It is recommended to appropriately reduce ATO dosage or adjust dosing interval, closely monitor ECG QTc interval changes
- **Concurrent Use of Cardiotoxic Drugs:** It is recommended to evaluate the necessity of concomitant medications and adjust the regimen if necessary to avoid additive cardiotoxicity
- **Low Secondary Methylation Capacity:** Patient may have weak arsenic metabolism capacity, it is recommended to strengthen cardiac monitoring frequency

VI. Precautions

- This report is generated by a machine learning model with an accuracy of approximately 85-90%, and misjudgment is possible.
- Prediction results should be combined with clinical manifestations, ECG, and echocardiography for comprehensive judgment.
- High-risk patients are recommended to strengthen ECG monitoring and adjust ATO dosage or suspend medication if necessary.
- Medium-risk patients should have regular ECG follow-up (every 3-7 days) and closely monitor QTc interval changes.
- Low-risk patients still require routine ECG monitoring and should not be completely complacent.
- Concurrent use of other cardiotoxic drugs (such as anthracyclines, fluoroquinolones) significantly increases risk.

VII. References

- 1. Zhu J, Chen Z, Lallemand-Breitenbach V, et al. How acute promyelocytic leukaemia revived arsenic. *Nat Rev Cancer*. 2002;2(9):705-713.
- 2. Unnikrishnan D, Dutcher JP, Varshneya N, et al. Torsades de pointes in 3 patients with leukemia treated with arsenic trioxide. *Blood*. 2001;97(5):1514-1516.
- 3. Barbey JT, Pezzullo JC, Soignet SL. Effect of arsenic trioxide on QT interval in patients with advanced malignancies. *J Clin Oncol*. 2003;21(19):3609-3615.
- 4. Shen Y, Shen ZX, Yan H, et al. Studies on the clinical efficacy and pharmacokinetics of low-dose arsenic trioxide in the treatment of relapsed acute promyelocytic leukemia. *Blood*. 1997;89(9):3354-3360.